

Institute of Actuaries of India

Subject SP4 – Pensions & Other Benefits

September 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

Experience of the DC scheme

Under a DC scheme the size of the pension benefits depends directly upon the experience of the scheme. Experience under the DC scheme may be very different from that assumed in the contribution rate of the DB system. [½]

For example, the new DC scheme members may have suffered poorer experience:

- poorer investment returns
- had higher annuity rates,
- members had higher salary increases
- incurred greater expenses

compared with that assumed. [1]

In particular, the investment returns may have declined over the years and the annuity rates have increased due to lower interest rates and lighter mortality.

Contribution rate

The contribution rate set up at the time of the transition of the DC system may have been inadequate to provide the same benefits under the DB system for the following reasons:

- The membership may have been ageing - The government may not have been using a prospective benefits method and therefore did not allow for ageing of these members [½]
- The DB contribution rate did not fully allow for future projected earnings [½]
- The DB contribution rate would depend upon the funding method and assumptions used to calculate the contribution rate. [½]
- It is possible that the assumptions used to calculate the contribution rate may be deliberately optimistic depending on the aims and objectives of the Government at that time. [½]
- For example, 25 years back, interest rates may have been significantly higher than what they are today. The rate assumed that the government will pay the pensions directly, which do not allow for annuities to be purchased from insurance companies [½]
- Assumptions did not allow for mortality improvement and significant lowering of interest rates [½]
- The contribution rate calculated will be based not only on the scheme retirement age but also on the probability that members will die, withdraw or retire early (due to ill health or otherwise). This may have led to some profits. For a DC system, there are no such profits. [½]

The DB contribution rate is a global rate, based on a weighted average of all the individual members' contribution rates. [½]

Where interest rates exceed salary increases, the cost of the pension provision increases as the age of the member increases. [½]

Hence the DB contribution rate will provide less than the cost of accrued benefits for older than average members, but younger members will be better off. [½]

The Chairman is comparing members who have retired recently - you would expect members retiring now may have been older than average at the time of the transition. Therefore, you would have expected the DB contribution rate to be insufficient to provide the benefits they would have received in the DB system. [½]

[Max 7]

ii)

DC provision in the organized sector could be via an occupational scheme or private provision. The Regulator may choose to impose different regulations in some areas for the two approaches. [½]

Areas where regulations may be imposed:

Operational – there may be restrictions on how such arrangements are set up, ie under trust or contract. There may be more protections if the scheme is set up under trust, eg assets held separately from the sponsor. [1]

Maximum contribution level (sponsor and/or employee) – there may be restrictions on the level of contributions, in particular if tax incentives are given. The maximum level of contributions may be age-related to reflect the cost of accrual increasing with age (provided investment return exceeds salary inflation). [1]

Minimum contribution level (sponsor and/or employee) – a minimum level of contributions could be prescribed to try to ensure adequate benefit provision. [½]

Taxation of contributions, investments, benefits – regulations may be imposed relating to the tax treatment of contributions, investment income and capital gains, pensions and any lump sum cash payment. [1]

Minimum level of benefits – may be required to target a minimum level of benefits, or schemes could be required to provide some form of defined benefit underpin. [½]

Type of investments – restrictions on investment choice could be imposed, eg limits on self-investment or the use of derivatives. [1]

Form of benefit provision – the form of benefit provision could be prescribed, eg require minimum % of pension for retirement below certain age. Regulations may allow some amount of the benefit to be commuted for cash. [1]

Age at which benefits may be taken – minimum and maximum ages of retirement may be prescribed. [½]

Provision of partner benefits – may require provision of a partner's pension if the member has a partner. [½]

Marketing of private benefit provision – regulation could be imposed on the quality of promotional material and the form of any illustrations. [½]

Level of charges – legislation could impose a maximum level of charges by the pension provider or asset managers [½]

Disclosure of information – legislation may outline the frequency and nature of disclosure of information. [½]

Advisors – requirement to take advice or for the advisor to be properly qualified and/or regulated. [½]

Regulation of finances of providers – eg solvency margins for insurance companies. [½]

[Max 9]

[16 Marks]

Solution 2:

i)

Possible impact on members' benefits of a switch to bonds

Provided the scheme is at all times sufficiently funded, there should not be any impact on the guaranteed benefits that would be provided to members

There may be enhanced security for accrued benefits particularly for liabilities matched by these assets, eg deferred and pensioner members.

However, if the switch to bonds leads to a lower return in the long term then the benefits will be more costly. There is the risk that the employer would no longer be able to afford to run the scheme leading to it being wound up, possibly with reduced benefits. These increased costs may mean that future benefits need to be reduced or discontinued, and also have a detrimental impact on active members' job prospects.

There may be an impact on discretionary benefit practice, e.g pension increases. If as a result of holding bonds, a lower return is achieved in the long term then members may not receive discretionary benefit increases, or they may be lower.

The switch to bonds may affect the amount paid as transfer values when deferred members take their benefits from the scheme. If the future benefits are now discounted at (lower) bond yields, then transfer value payments will increase.

Similarly, the switch to bonds may affect option terms, e.g. more generous commutation factors.

[Max 5]

ii) Factors to consider when choosing the bond investments

The trustees will wish to maximise net return subject to an acceptable level of risk.

An important consideration is the degree to which risk is reduced through matching liabilities by nature, term, currency and uncertainty.

The liabilities for actives will probably be real in nature (for a final salary scheme linked to salary growth). This suggests holding index-linked bonds.

The nature of the liabilities for deferred and current pensioners depends upon the nature of deferred and current pension increases, they could be fixed or index-linked or linked to inflation and capped. Hence fixed-interest and/or index-linked bonds may be required.

We should consider whether and how the scheme's liabilities are likely to change in the future, *eg* is the scheme closed to new entrants? Is the fund expanding or contracting?

The scheme's current funding level should be considered.

The size of the fund will also be important and in particular relative to the size and financial strength of the employer.

Bonds of an appropriate term need to be chosen. This is possible for current pensioners, however there will be reinvestment risk for deferred pensioners and actives as bonds of a long enough term may not be available.

Domestic bonds will be the most suitable since the liabilities will be in the domestic currency.

Guaranteed benefit payments must be made so the bonds chosen must have a low risk of default, this suggests government bonds and high-quality corporate stock.

The trustees will wish to maximise net (of expense and tax) return, they will look for bonds which are "cheap", in particular given the tax treatment of the scheme which may differ from that of the average investor. This may also lead to a preference for corporate bonds over government bonds as the former usually have higher returns.

The trustees will seek to diversify within the bond holding in order to reduce risk.

The trustees will look at the scheme's maturity and hence cashflow position which will dictate whether low, medium or high coupon bonds are required.

Holding high coupon bonds could have advantages if most other investors prefer growth over income.

[Max 6]

iii) Other issues for deciding on investment objective and asset allocation

The risk appetite of the employer sponsoring the scheme:

- a low-risk investment strategy will give stability to the contribution rate over time
- a higher risk strategy will lead to a more volatile contribution rate but should have lower costs in the long term.

The risk appetite of the trustees is also important.

The trustees will consider the strength of the employer, and the size of the employer relative to the scheme in determining an acceptable level of risk.

Will both the employer and trustees be prepared to accept under-performance due to poor market conditions?

The level of understanding and expertise of the trustees and the need for additional advice needs to be considered.

The scheme's current funding level – the higher the funding level the greater the scope to hold riskier investments.

The level of insurance used – if protection benefits have been insured there is greater scope to hold riskier investments.

Whether pension benefits are to be paid from the fund (likely with a large scheme) or bought out with an insurance company affects the term and nature of the liabilities and hence the investments held for matching purposes.

The scheme's intended discretionary policy.

Regulation, for example:

- the statutory valuation basis, *eg* if liabilities have to be valued using a discount rate based on bond yields then holding bonds increases the stability of the funding level
- whether there are restrictions (or requirements) for holdings of certain asset classes or individual stock.

The current investment holding and market conditions, for example is it a good time to make a large switch? Are there marketability and liquidity concerns?

The need for diversification within and across asset sectors.

Any legislation governing investment holdings, *eg* minimum / maximum holdings in certain asset classes.

The holdings of similar competitor schemes will be examined, has there been a general trend towards a greater holding of bonds?

The trustees may want to arrange for an asset liability modelling exercise to be carried out in order to identify optimal investment strategies.

[Max 7]
[18 Marks]

Solution 3:

i)

General considerations before introducing flex scheme:

Competition: Do other firms in the industry already offer flexibility with benefit entitlements? [½]

Legal: Do the proposed changes have an impact on the employees' contracts of employment [½]

Legislation: Is there any legislation relevant to the proposed benefit change? Are any restrictions set out in law? [½]

Financial: What are the financial consequences of the proposed scheme for the company and the employees? Do changes to salary have any knock-on effects on tax, take-home salary or entitlement to State benefits? [½]

For example, could this lead to anti-selection? What would be a fair price to exchange these benefits? [½]

How flex scheme may meet client's objectives

The employer has many objectives. Two important objectives are:

- to provide benefits at a minimum cost [½]
- to provide competitive benefits to recruit and retain staff. [½]

These objectives are contradictory. Any solution will be a compromise that provides the best benefits at an acceptable cost. [½]

The actual cost of benefits is less important than the perceived value of benefits amongst staff. [½]

The flex option acknowledges that different members of staff have different needs, which change over working lifetime of employees. For example, current holiday arrangements may not be sufficiently flexible to cope with a wide variety of personal circumstances. [½]

Without a flexible approach to benefits, potential recruits may have been deterred from joining if other employers offer more of some of these benefits. [½]

A flexible approach will also help to retain staff who value such benefits highly and would otherwise be tempted to change employer to increase their entitlement. [½]

If decisions about unpaid leave have previously been left to individual managers, staff may have been treated unequally, resulting in a disgruntled workforce. [½]

The benefit may be used to assist in manpower planning. [½]

For example, if the management consider that the firm is currently overstaffed then individuals could be encouraged to increase their holiday entitlements in the short term. [½]

This would avoid redundancies. [½]

[Max 8]

ii)

*Risks and problems***Pricing**

- Consideration should be given to obtain a fair price for the exchange of benefits
- The pricing of the option must be simple so that it can be understood by the members of staff and is perceived as fair
- What are the assumptions that should be used?
- Should be cost neutral for the employer
- For example, will the price for buying leaves be the same for selling leaves?

Selection

- The flex scheme could lead to selection - Different groups of employees may have different preferences, eg younger employees may prefer to take holiday in lieu of salary. Even if the overall level of holiday within the firm is unchanged, there may be staff shortages in certain areas. [½]
- Selection is particularly concerning when offering cash for risk benefits / insurance:
 - e.g. younger employees may not opt for medical insurance if they think they are unlikely to use it
 - Older employees may buy additional medical and life insurance
- This will raise the overall cost of medical and life claims.
- If too many employees opt to increase holiday, the firm may not be able to cope with the workload. In this case, costs could increase if the firm is forced to employ temporary staff. [½]
- The total cost of benefits will increase in respect of all employees who regularly do not use their full holiday entitlement. These employees can now flex down to the appropriate level of holiday and take the value of any unused holiday allowance as cash. [½]

Core benefits

- Consider if there is a minimum level of benefits below which employees cannot fall
- Encouraging staff to reduce their annual holiday entitlement has potential problems:
 - productivity may decrease if employees do not have a break from work to recharge [½]
 - employees may take more paid sick leave. [½]
- There could be legal minimums for annual leaves
- Similarly, there should be a minimum level of coverage for life and medical insurance

Other

- Special consideration may be required for part-timers and shift workers who have different work patterns and leave opportunities [½]
- Changes to salary may have a knock-on effect on pensions and other benefits that are based on salary. [½]
- Changes of salary and flex administration will potentially produce a lot of extra work for the payroll and benefits department. Can the admin system and staff cope? [½]

[Max 8]**[16 Marks]**

Solution 4:

i)

- Immediate annuities purchased by a single premium could remove a liability to a current pensioner
- The immediate annuities may be increasing at a fixed rate or increasing in line with a given index. They may continue after death to a surviving spouse, at either the full or a reduced amount
- Deferred annuities could extinguish a scheme's liability to a member with an entitlement to a deferred pension
- The annuity may have to increase in both deferment and payment, either at a fixed rate or at a rate linked to an inflation index.
- Schemes can have complexities such as differing levels of increases - is it possible to match the nature of the scheme with an annuity?
- Longevity swaps could be purchased whereby the scheme pays a guaranteed level of payments to the insurance company in return for the insurance company to pay the actual payments that would be required to pay the members (i.e. the company takes the risk over and above the agreed level of payments).

[3]

ii) As required the main advantage of these products is that

- They reduce the longevity risk
- This would reduce long term cost uncertainty for the scheme and so reduce reliance on sponsoring employer's support.
- For the immediate and deferred annuities it also removes the inflation and investment risk, depending on the financial product used
- Longevity swaps still leave the scheme with the opportunity of making investment profits

The disadvantages of the immediate and deferred annuities are that:

- It removes the upside longevity, inflation and investment risks – however given the scheme wants to remove longevity risk it is only the 2 latter risks that are a disadvantage in this case
- The purchase price has to cover insurance company expenses and a contribution to its profit which may make the product relatively expensive
- Dealing with issues such as future addition of discretionary pension increases could become complicated depending on whether they remove all the risks when transferring to an insurance company (either way these issues could be complicated)
- Need to have liquid cash available
- Longevity swaps still means the scheme has inflation and investment risk and will need to decide whether to cashflow match the payments to the insurance company or maintain the investment risk with the hope that they outperform
- The market for longevity swaps may be in its infancy and thus hard to find an insurance company to enter into the swap with

- There is counterparty risk with any such product.
- Data issues. There data may well be old and there may have been changes to benefits so tying down benefits exactly may be a difficult exercise

[4]

iii) a. Merits

- One off increase in the scheme will reduce the tail risk where increasing longevity will be coupled with continual increasing benefits
- Still have the longevity risk for the individual member and there will be a cross over point where the scheme would win/lose from this transaction

Transfer value

- If the member took this option then the scheme will have reduced the longevity risk, and indeed all investment/etc. Risks and the individual will have removed their longevity risk
- But the scheme needs to pay out the enhanced transfer value now, and it will pay out more than the benefits would have cost in the case of a member who does not live longer than average

In both cases there are anti-selection issues, whereby members who are in poor health take the option.

- E.g. for the transfer value offer, because they can buy an enhanced annuity from an insurance company whereby they could get an increased pension,
- or for the pension uplift because they are just giving up future increases that they wouldn't have lived to receive.
- If the scheme is left with members who are in good health (and can't get a better deal elsewhere), it ends up with a worse longevity risk than when they started

The scheme will need to consider if doing either of these makes any further de-risking options more expensive in the future – e.g. will life insurance companies think that the scheme has been selected against and hence be more expensive by assuming lighter mortality for anyone who has not taken the offers?

How much will it cost to run the offers, including communications and administration?

And will the changes affect the ongoing admin of the benefits (e.g. for the uplift option will having some members having increases and some not cause the admin teams problems)

The scheme will need to consider how it funds the transfer value option and to a lesser extent the uplift option, what assets will it sell and does it have liquid enough assets to fund if the offer is successful

[3]

b. Factors

For both options the scheme will need to decide on how generous to be – this will probably depend on how paternalistic the scheme feels to its members

In both cases, the take-up will depend on how generous the offer is. There is an inverse relationship with generosity and take up, The smaller the uplift to the pension now (or enhancement to TV) the more benefit the scheme will see but probably a lower take up from the membership, the higher the uplift/enhancement the higher the take up would be BUT the lower the benefit to the scheme

The scope for enhancing the TV will depend on how prudent/otherwise the TV basis currently is

Targeting certain risk groups. The scheme will need to decide whether it asks for medical information before deciding the enhancement to offer, to reduce anti-selection risk, or whether to offer to just joint life policies – this would probably seem to be unfair and not filled in correctly if at all. If the members who have not retired are still working for the sponsoring company then the scheme will need to consider the impact on the members/employees if the deal offered is not seen to be fair or reduces their benefit security

Will need to know whether the company is prepared to put money into the enhancements to get a higher take up rate such that more longevity risk is removed

Funding level and whether the company can afford to will influence the generosity

Legislation may require that the members are able to get financial advice should they require it (such that the members know what they are giving up – e.g. benefits of increases). And therefore would want to take account of what the advice might be when designing the offers

For the transfer value option will there be an option to join a defined contribution scheme (either run by the company/scheme or another)

Will need to consider the tax issues before deciding on the structure e.g. if members accepting the offers might incur higher tax.

With both options the scheme will need to be professional and consider regulatory and reputational issues

The scheme should consider variations to the designs: for example to pay a guaranteed cash sum rather than a pension uplift.

[6]

iv) Longevity swaps and longevity bonds

A major risk affecting defined benefit pension schemes is longevity risk. This can be mitigated by investing in longevity bonds or swaps.

Longevity bonds work where the coupon payments are linked to the mortality experience of a particular set of lives, such as a specific age group in the national population. Coupon payments will reduce over time in line with the mortality experience of that population.

Larger schemes may be able to arrange a longevity swap with an insurance company where the payments are linked to the mortality experience of the scheme's membership rather than the national population.

If longevity is higher than expected, the coupon payments received will be higher, thus hedging the longevity risks of the pension scheme.

The hedge is not perfect if the population underlying the bond is not the same as that in the pension scheme. For example, if the pension scheme members live longer on average than the population being tracked. The hedge will also not be perfect if investment returns and / or pension increases are at a different rate than expected.

The market for longevity bonds is currently very small.

[4]

v) The commonly used sources of information available to help assess a sponsor's credit quality are:

- *Financial metrics*
- *Implied market default risk*
- *Credit rating*
- *Merton-type credit risk models*
- *Quantitatively derived credit risk*
- *Independent business review (IBR)*

[3]

vi) *When sponsor covenant should be considered*

In almost all cases an assessment of the sponsor covenant could be considered to be fundamental to the trustees when considering actuarial advice. However, there are a few exceptional circumstances where this is not the case.

The sponsor's covenant is not an important issue if either the reliance of the scheme on the sponsor covenant can be treated as certain or if it is clear that the scheme will not need to rely on the future support of the sponsor. Examples would include:

The scheme is very well funded, therefore no reliance is placed on the sponsor.

A measure of this might be that the scheme has more than sufficient assets to secure all the scheme's accrued liabilities on a buy-out basis (*ie* with an insurance company), and the trustees have the right to stop the sponsor adding to the liabilities.

The sponsor covenant is strong enough to be deemed as certain One example may be a scheme that is backed by the State, *eg* a public sector scheme.

The sponsor covenant is so weak as to be deemed nil, and hence will not affect security

The sponsor has no further liability, an example of this would be if the sponsor has met all its legislative requirements towards the scheme, *eg* paid any debt on the sponsor due on discontinuing the scheme.

In all other circumstances, sponsor covenant should be taken into account.

[4]

[27 Marks]

Solution 5:

i)

- A Lump sum in advance when individual commences employment;
- B Regular contributions throughout the period of their employment;
- C Lump sum at the time of their retirement
- D Pay-as-you-go as benefit payments is due.

Surplus/deficit does not arise under D.

Generally, surplus might be dealt with by reducing future contributions, but there is no scope for this under A, and only limited scope under B and C surplus may arise after retirement. An alternative is to take a refund of surplus.

Generally, deficits are dealt with by the employer paying additional contributions. This may happen after the employer thinks he has finished paying, and a significant unexpected contribution requirement may create problems for the employer, especially if the KMPs are no longer employed. Another option is to adjust benefits, but that defeats the objective. There may be legislative constraints and/or tax implications.

[5]

ii)

Capital requirements can be considered in terms of economic capital or regulatory capital.

Regulatory capital will depend on what legislation applies (if any) and on what accounting rules apply.

In economic terms, the company is taking on an obligation which reduces its available capital; the obligation is the value of the benefits and expenses that will arise. There are a variety of ways to value this but it is important to allow for the uncertainty: the risk that the actual obligation exceeds the expected cost.

The company could aim to secure this with an insurance contract at retirement

- consider the expected premium - or even look at the cost of securing the benefit now.

Even if it is not intended to secure the benefit with insurance, the cost of doing so is a good measure of the cost of the capital required to provide it.

The company needs to ascertain whether the obligation accrues immediately or over the course of the KMPs career. The latter is preferable so that the capital cost can be met from profits over the period, which is the period that the company benefits from the KMPs work

Also consider the benefits payable if any of the KMPs dies or leaves before age 60.

There may be currency risk consider if that could be hedged.

[7]

iii)

Establish exactly the target benefits as both KMPs have different risk appetite

... their dependant s benefits, pension increases in payment

... is the target pension really a fixed amount at age 60, or indexed

Establish the contribution pattern

... regular level contributions or increasing with salary.

how much is affordable

If anyone of them died, or retired early or late, how much would be required in the fund.

Do they intend to purchase an annuity at retirement with some or their entire fund

Establish attitude to risk, given the uncertainty of outcomes.

Are there particular risks to be avoided or that the individual (depends on the risk appetite of both the KMPs) is prepared to accept.

... do the KMPs prefer to aim high or low?

... is there scope to adjust contributions in response to changing circumstances?

Establish how he intends to invest his fund and whether that would change over the period to retirement.

Obtain information on his health and life expectancy.

Establish his tax position on pension and contributions

[7]

iv)

Remind the individual KMPs of the inputs they gave

State that the answer will depend on actual experience so you need to make assumptions, and set out the key assumptions made.

Illustrate the variation by using a range of scenarios or stochastic analysis.

Point out that the uncertainty will be affected by his investment choice, and possible changes to tax, legislation, etc.

Suggest that there is regular review to see whether the contribution plan continues to meet his objectives.

[4]

[23 Marks]
